

PERVEZ ALI

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Fulbright Scholar
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RESEARCH INTERESTS

Design and analysis of Langevin-type Markov chain Monte Carlo algorithms, especially nonreversible dynamics and sampling on constrained domains; nonasymptotic convergence guarantees and large deviations analysis of samplers, paired with careful numerical studies; stochastic optimization and sampling for Bayesian inference and generative modeling.

EDUCATION

Florida State University <i>Ph.D. in Mathematics</i>	Expected June 2027 Tallahassee, FL, USA
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- Advisor: Lingjiong Zhu
- Dissertation: *Non-Reversible Langevin Dynamics for Unconstrained and Constrained Sampling*

Sukkur IBA University <i>M.S. in Mathematics</i>	2019 Sukkur, Pakistan
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- Focus: probability, stochastic processes, and mathematical finance (option pricing)

University of Sindh <i>B.S. in Mathematics</i>	2016 Jamshoro, Pakistan
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PUBLICATIONS AND PREPRINTS

- [1] **Pervez Ali**, Weihao Dong, and Xiaoyu Wang. *Penalized nonreversible Langevin for constrained sampling*. Preprint, [arXiv:2609.25381](https://arxiv.org/abs/2609.25381), 2026.
- [2] Nian Yao, **Pervez Ali**, Xihua Tao, and Lingjiong Zhu. *Accelerating Langevin Monte Carlo sampling: A large deviations analysis*. Preprint, [arXiv:2503.19066](https://arxiv.org/abs/2503.19066), 2025.
- [3] Javed Hussain and **Pervez Ali**. *On modelling and pricing weather derivatives driven by nonlinear Brownian motion*. *International Journal of Analysis and Applications* **19**(1):29–46, 2021.

TALKS

- *On modelling and pricing weather derivatives driven by nonlinear Brownian motion*. Jan 2020
3rd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET),
Sukkur IBA University

CONFERENCES AND SCHOOLS ATTENDED

- Southeast Applied and Computational Math Student Workshop, Georgia Tech, Atlanta, GA Apr 2024
- Hausdorff School: Formal Mathematics and Computer-Assisted Proving, University of Bonn Sep 2023
- Winter School on Risk, Stochastic, Finance and Insurance, Sukkur IBA University Dec 2018
- 2nd iCoMET, Sukkur IBA University Mar 2018

TEACHING EXPERIENCE

Florida State University

Graduate Teaching Assistant and Instructor

Aug 2022 – present

Tallahassee, FL, USA

- **Instructor:** Calculus I (Spring 2025, Summer 2025), Calculus II (Fall 2025, Spring 2026), Calculus III (Summer 2026, Fall 2026).
- **Teaching assistant:** Calculus, Trigonometry, College Algebra; tutor, Math Resource Center.

Sukkur IBA University

Lecturer in Mathematics

Aug 2017 – Jul 2022

Sukkur, Pakistan

- **Mathematics:** Calculus I–III, Linear Algebra, Ordinary Differential Equations, Scientific Computing.
- **Probability and statistics:** Probability, Statistics, Inferential Statistics.
- **Business and actuarial:** Business Mathematics, Business Statistics, Actuarial Mathematics.

FELLOWSHIPS AND AWARDS

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| • Educator of the Year Award | 2023 |
| • Travel grant to attend the Hausdorff School, University of Bonn | 2023 |
| • Fulbright Scholarship for Ph.D. study in the United States | 2022 |
| • Departmental Merit Scholarship, Sukkur IBA University | 2019 |
| • Prime Minister's Scholarship for M.S. study (fully funded) | 2018 |
| • Student of the Year in Mathematics, University of Sindh (awarded a laptop) | 2015 |

SERVICE, OUTREACH, AND MENTORING

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| • Vice President, SIAM Student Chapter, Florida State University | 2023 – 2024 |
| • Fulbright outreach: led seven information sessions on the Fulbright program at three institutions. | |
| • Mentored undergraduate projects at Florida State University and Sukkur IBA University. | |

COMPUTING AND LANGUAGES

Computing: Python and MATLAB.

Languages: English (fluent), Urdu (native), Sindhi (native).